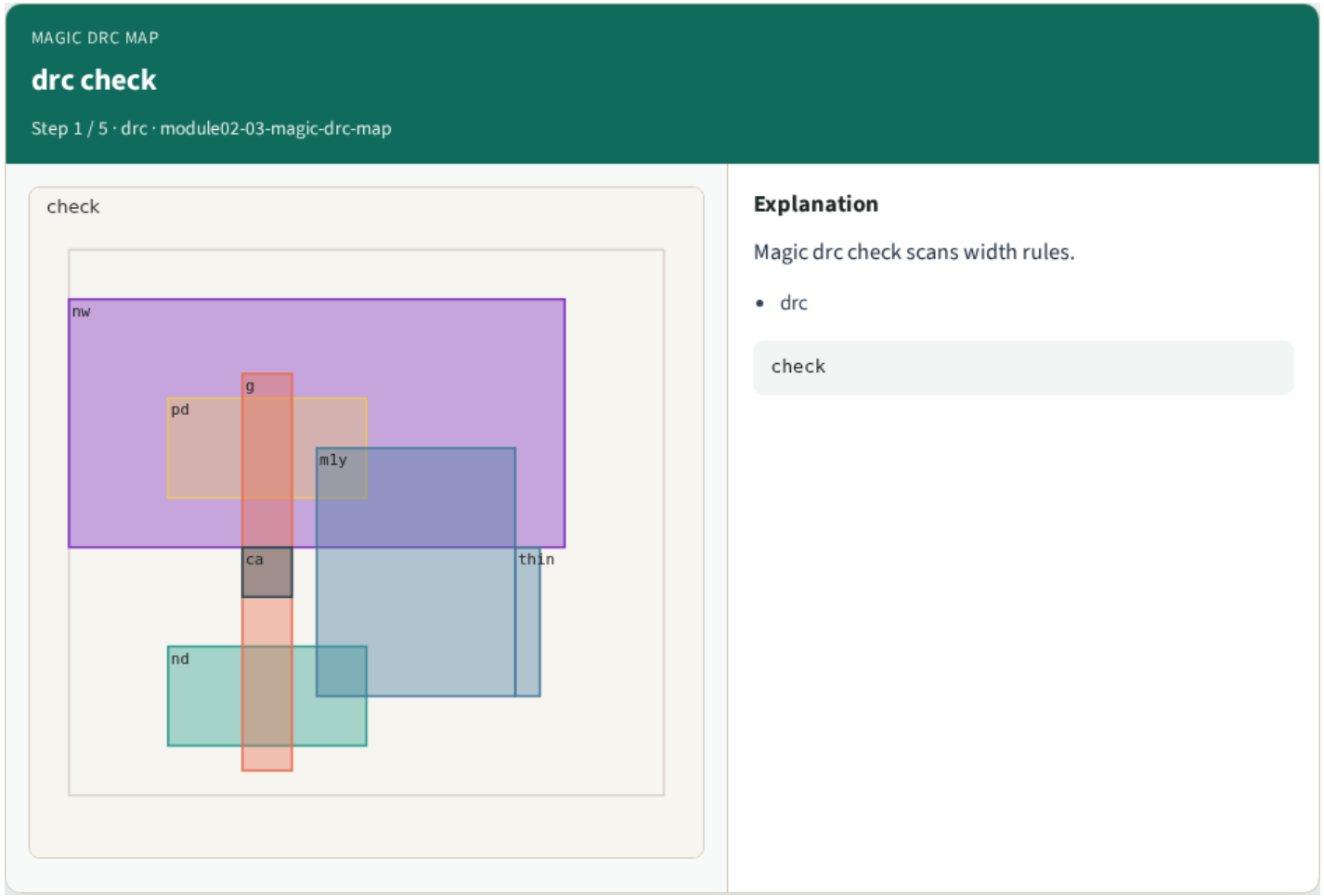
# Why Magic

Run DRC width check; find violation on thin metal1

# The idea

- GOLDEN: drc_error_count=1 on thin

# drc check



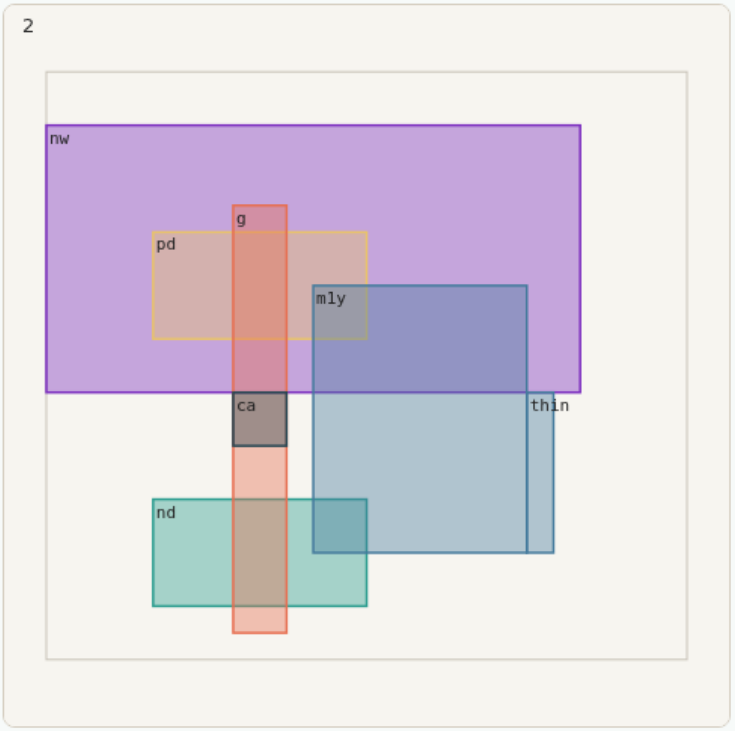

drc check

# min_width

MAGIC DRC MAP

**min_width**

Step 2 / 5 · rule · module02-03-magic-drc-map




**Explanation**

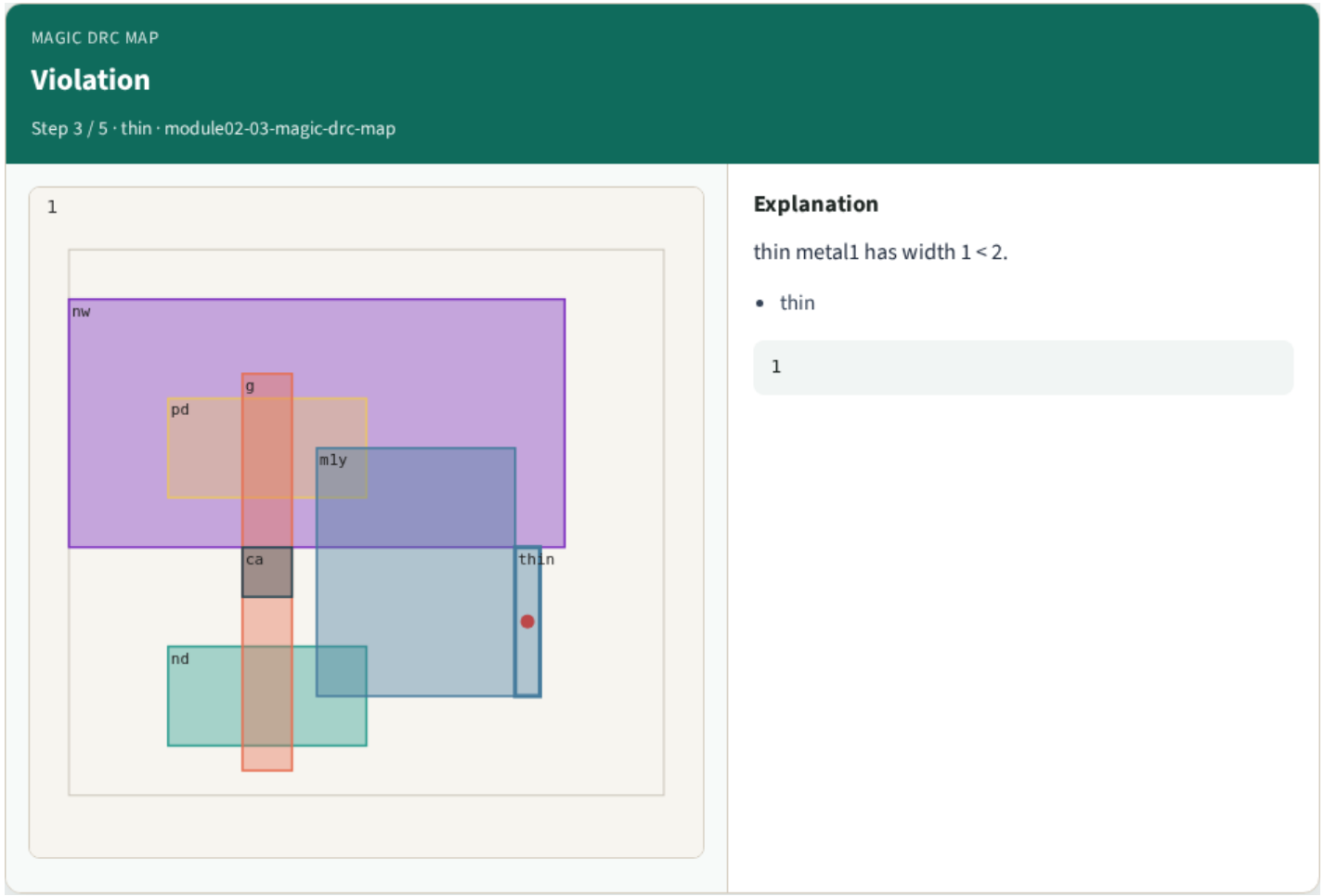
metal1 min_width=2 on this tech.

- metal1

2

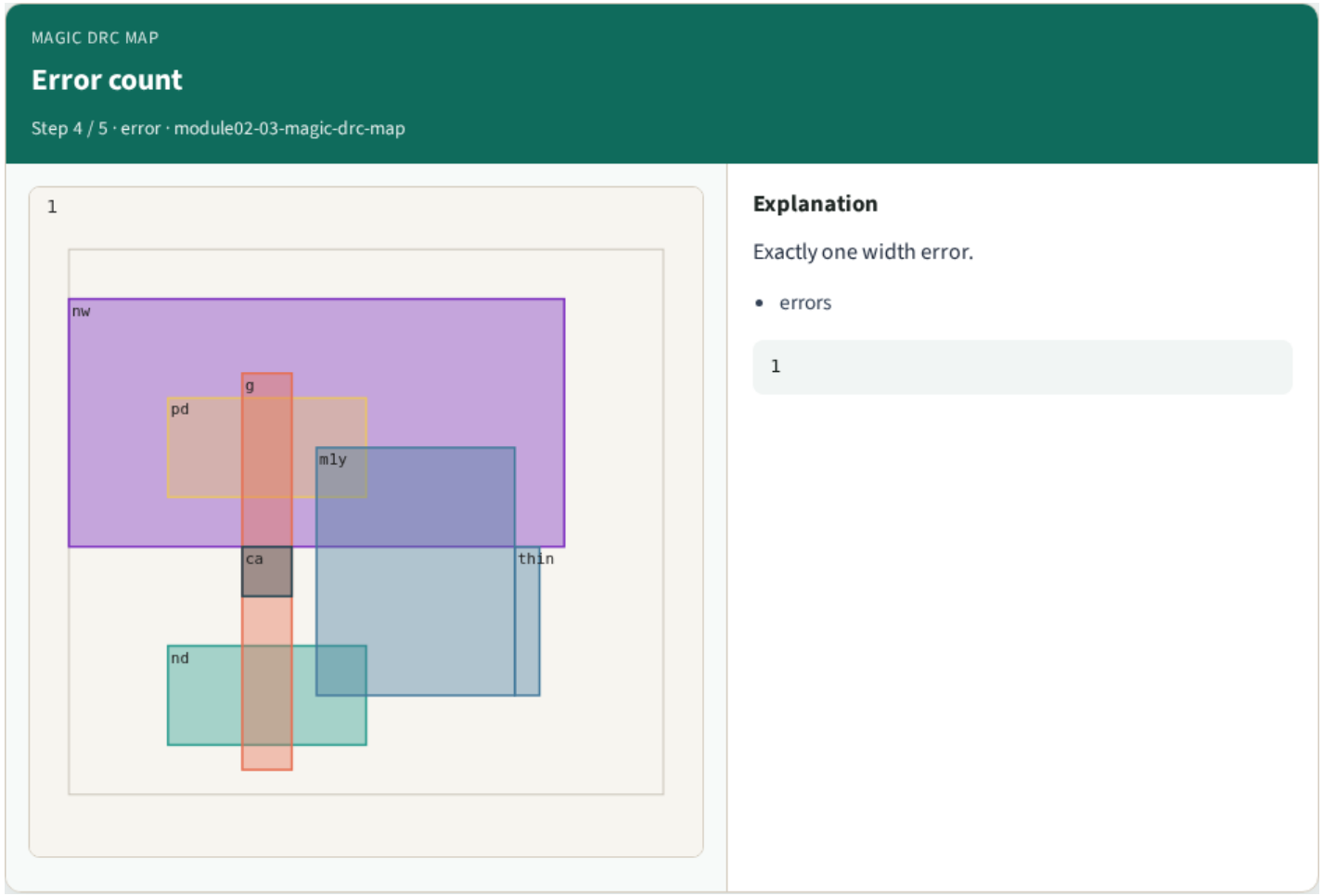
min_width

# Violation




Violation

# Error count




Error count

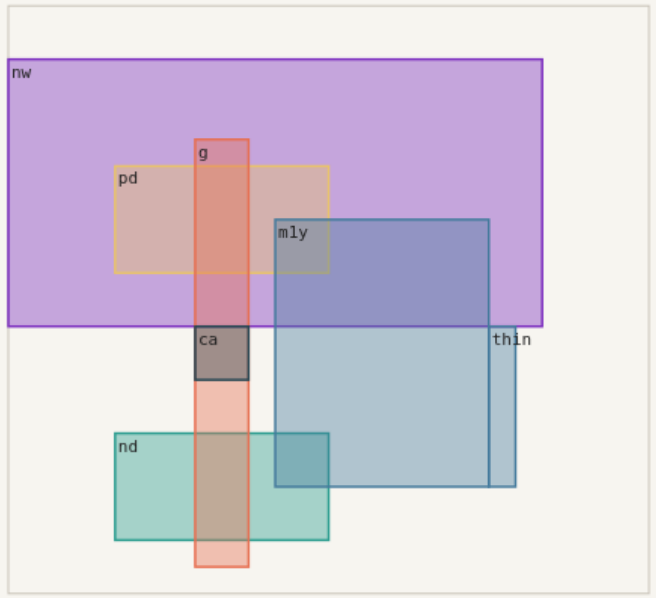
# Golden

MAGIC DRC MAP

**Golden**

Step 5 / 5 · golden · module02-03-magic-drc-map

1/thin




**Explanation**

GOLDEN: drc_error_count=1 on thin.

- GOLDEN

```
1/thin
```

Golden

# Browser lab track

- Open **magic-drc-map**
- Run the stage buttons and Check the goldens

# Implement track

- From common/, run solvers.py and confirm MAGIC_PARTA_OK

# Pitfalls

- Mixing KLayout viewer shortcuts with Magic paint commands
- Forgetting min_width when reading DRC errors

# Try these

- From the common folder, run the Magic map self-test

# Try these — try these

```
cd courses/learn_magic/common
python3 solvers.py
```

# Pseudocode

- Here is the check in plain steps

# Pseudocode — try these

```
drc_error_count=1 on thin.
ok if goldens match
```

# Next

- Complete the quiz for this part
- Continue the Magic Part A map, then the real CLI track